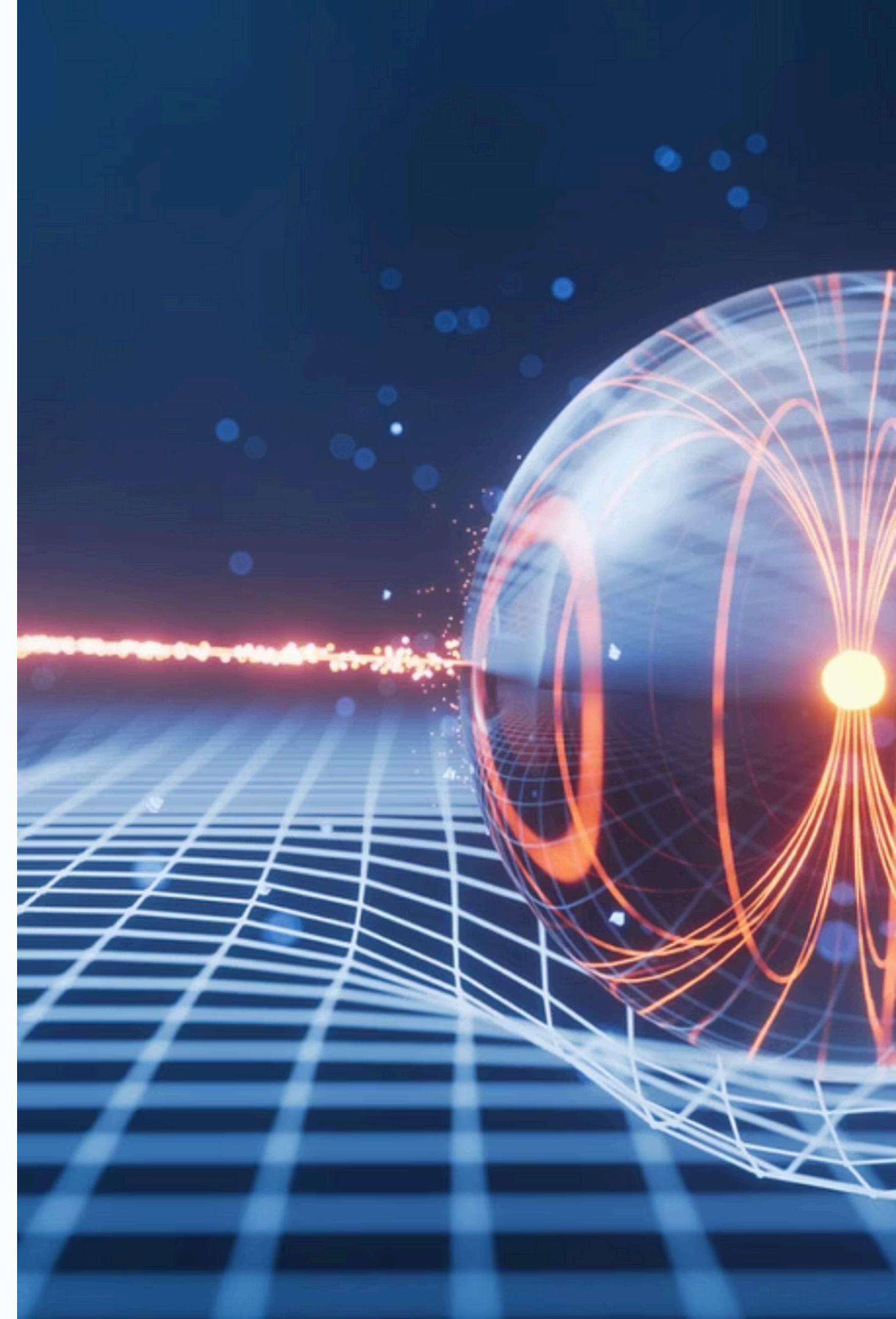


Quantum Methods for Location Tracking

By Mohamed Khalil Brik

SID: 900225905

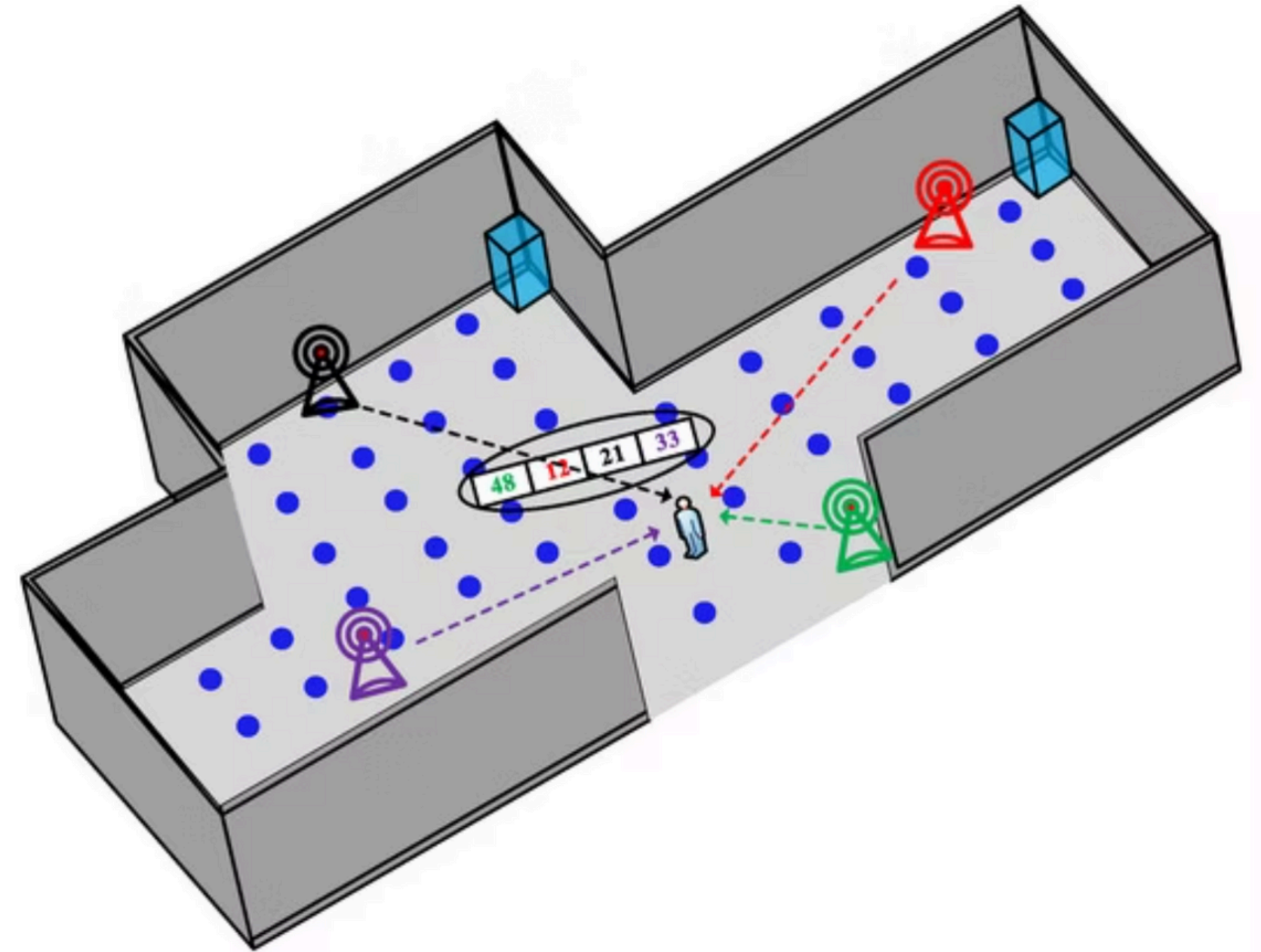
Supervisor: Dr Moustafa Youssef



Why am I here? > The Indoor Localization Challenge

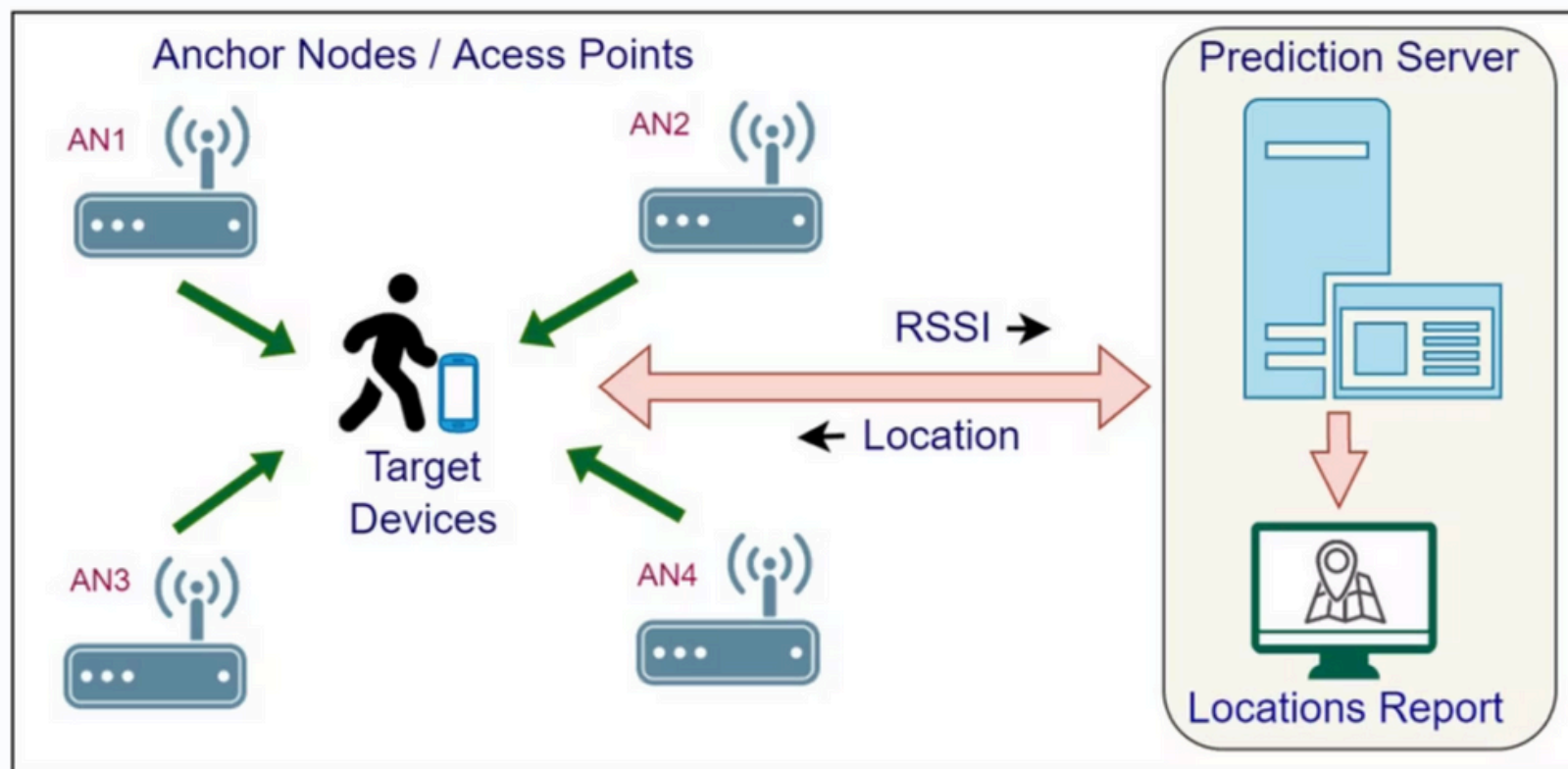
- GPS degrades sharply indoors due to signal attenuation, multipath propagation, and lack of clear line-of-sight.
- People spend roughly 90% of their time indoors.
- Accurate indoor localization is the base for:
 - Incident management
 - Healthcare operations : emergency response
 - Retail environments
 - Industrial automation and logistics
- WiFi fingerprinting is a proven indoor method.
- Fingerprinting performance depends on AP quality, placement, and selection → affect accuracy, and computational cost.

→ GPS is not enough indoors, the majority of human activity happens inside, and mission-critical applications require dependable, precise indoor positioning.



Zhang, H., Zhang, Z., Gao, N., Xiao, Y., Meng, Z., & Li, Z. (2020). Cost-Effective Wearable Indoor Localization and Motion Analysis via the Integration of UWB and IMU. *Sensors*, 20(2), 344. <https://doi.org/10.3390/s20020344>

What is the current solution? > WiFi Fingerprinting



WiFi fingerprinting remains a leading technique for indoor localization. However, its accuracy, robustness, and computational efficiency are heavily influenced by the selection and deployment of Access Points (APs).

Non-optimal AP placement can lead to:

- Reduced localization accuracy.
- Increased signal interference.
- Higher infrastructure costs.
- Complex maintenance and calibration.

Optimizing AP configuration is critical for scalable and reliable indoor positioning systems.

Varma, P. S., & Anand, V. (2022). Intelligent scanning period dilation based Wi-Fi fingerprinting for energy efficient indoor positioning in IoT applications. *The Journal of Supercomputing*, 79(6), 7736–7761.

<https://doi.org/10.1007/s11227-022-04980-9>

Research Question

How can we place n APs in k candidate locations to maximize accuracy in a 3D Environment?

What we learned? > Literature Review

Classical Localization

GNSS (Global Navigation Satellite System) is effective outdoors, but indoors, WiFi fingerprinting remains dominant. Variants include deterministic, probabilistic, ML/deep learning, federated learning, and reinforcement learning approaches, each addressing robustness and scalability challenges.

Quantum Localization

Emerging techniques like QLoc and quantum fingerprinting encode RSSI into quantum states for superposition-based similarity. These methods promise logarithmic complexity under idealized assumptions but are currently limited by data loading and noise.

Optimization Approaches

AP selection has been explored through greedy algorithms, convex/non-convex relaxations, graph-based solutions, and RL. Quantum approaches (QA, QAOA, Grover's algorithm) encode these problems as QUBO/Ising models to efficiently search combinatorial spaces.

What is our solution? > Quantum Optimization

Quantum Annealing

Utilizing quantum annealing to solve complex optimization problems, particularly those involving discrete choices.

AP Selection & Placement

Formulating AP selection/placement as a Quadratic Unconstrained Binary Optimization (QUBO) problem.

Enhanced Performance

Maximizing accuracy and efficiency by intelligently choosing or positioning APs under various constraints.

What we will achieve? > Project Idea & Objectives

QUBO Definition

Define a clear QUBO for AP selection/placement with explicit cost and constraint terms, where binary variables indicate AP choices and quadratic terms encode informativeness, redundancy, and penalties for cardinality and coverage.

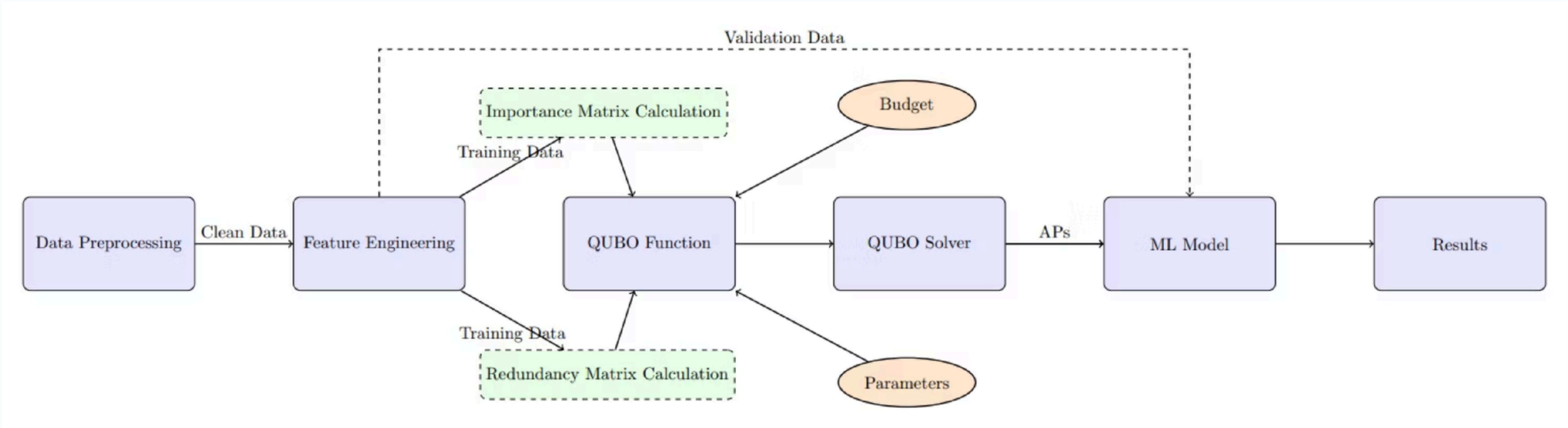
Comparative Analysis

Solve the QUBO on a quantum annealer and benchmark against classical selectors (greedy, convex, RL) across metrics like accuracy, robustness, and runtime.

Evaluated Pipeline & Artifact

Deliver a comprehensively evaluated pipeline and artifact suitable for research dissemination and potential future deployment, aiming to achieve full-set accuracy with a minimal AP subset and demonstrate runtime/energy benefits over classical methods.

How we will achieve it? > Our Architecture



This graph is developed using Latex



What we expect? > Expected Advantages of Quantum Optimization



Efficiency Gains

Prior QA-based AP selection achieved full-set accuracy with <14% of candidate locations, demonstrating significant efficiency without sacrificing accuracy.



Flexible Modeling

QUBO modeling seamlessly combines accuracy, redundancy, and budget constraints, allowing for adaptable problem definitions.



Accelerated Search

Hardware annealing can navigate complex, rugged solution landscapes more rapidly than many classical heuristics, especially as problem scale increases.



Hybrid Stability

Hybrid quantum-classical loops stabilize results, reduce required annealing shots, and enable adaptive solutions to dynamic AP visibility changes.

What we achieved

- ✓ Accessed and processed a comprehensive data set with 520 Access Points and 19,937 fingerprints, spanning 5 floors across 3 buildings.
- ✓ Developed a robust data pipeline, encompassing crucial steps like data cleaning and feature engineering.
- ✓ Successfully formulated the Access Point selection problem as a Quadratic Unconstrained Binary Optimization (QUBO) model.
- ✓ Calculated the Importance matrix using mutual information to identify key Access Points.
- ✓ Quantified redundancy among Access Points using Pearson correlation.
- ✓ Solved the formulated QUBO problem using classical Simulated Annealing techniques.
- ✓ Implemented and solved the QUBO problem using Simulated Quantum Annealing via the Openjij library.



Table 1: Performance Comparison of OpenJij vs Simulated Annealing for Indoor WiFi Localization

Performance Metric	OpenJij	Simulated Annealing
Localization Accuracy		
Training Real Error (m)	3.32	3.38
Validation Real Error (m)	8.65	9.18
Floor Accuracy (%)	89.2	86.2
90th Percentile Error (m)	17.65	20.15
Model Performance		
Training Normalized Error	0.0408	0.0485
Validation Normalized Error	0.1109	0.1279
Overfitting Ratio	2.72	2.64
Solver Performance		
Selected Access Points	50	50
Feature Reduction (%)	90.4	90.4
Solver Runtime (s)	137.13	63.22

What is next? > Time Plan

Semester 1

1. Problem Definition & Scope (Week 1 -Week 2)

- Finalize thesis problem scope.
- Define QUBO formulation of the AP placement problem in 3D environments.
- Identify datasets and simulation environments for AP placement.
- Establish evaluation metrics (coverage, accuracy, cost, etc.).

2. Baseline Implementations (Week3)

- Implement classical baseline: Simulated Annealing.
- Implement quantum-inspired baseline: Simulated Quantum Annealing.

3. Quantum Hardware Mapping (Week 4)

- Map AP placement QUBO to D-Wave quantum annealers.
- Conduct preliminary experiments on D-Wave.

4. Comparative Evaluation (Week 5 - Week 6)

- Compare results across classical vs. quantum-inspired vs. quantum hardware approaches.
- Analyze performance using chosen metrics.
- Tune parameters for annealing methods.
- Conduct evaluation of approach.

5. Documentation & Dissemination (Week 7 - Week 10)

- Draft design + implementation plan.
- Write preliminary thesis sections: problem scope, methodology, early results.
- Draft the paper.

Extended Exploration (time-permitting)

Investigate **quantum approaches in location tracking**:

- Quantum fingerprinting.
- Quantum ML for fingerprinting.

Explore **related optimization problems**:

- Signal source weighting.
- Satellite selection.
- Cell tower selection.

- Formulate each as a QUBO and run experiments.

Semester 2

- Finish Paper Writing
- Finish Thesis writing
- Finish Experiments

What is our impact? > Empowerment Outcomes

Outcome 1: The primary outcome is a comprehensive research paper, supported by rigorous literature review, formal QUBO modeling, quantum/classical comparative experiments, and analyzable results suitable for submission to top localization or quantum systems venues.

Outcome 2: Optionally, a prototype package for academic/industrial evaluation (deployment plan and scripts) may be developed if time permits, building on the validated pipeline and tooling described.

Thank You!

This concludes our presentation on:

Quantum Methods for Location Tracking

Presented by: **Mohamed Khalil Brik**

Supervised by: **Dr. Moustafa Youssef**

We appreciate your time and attention.

Any Questions?